VANISHING OF HIGHER EXTS SPLITS A BOUNDED COMPLEX INTO ITS COHOMOLOGY OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 9235-9252, following the statement at 9227-9233.

Editorial note. No omissions: the induction step displays the Ext direct sum identification.

Allowed standard prerequisites (editorial).

- Truncation triangle. remark-truncation-distinguished-triangle (same chapter, cited at ~9240). Provides tau_{<=b-1}K -> K -> H^b(K)[-b] -> ...
- Split triangles. lemma-split (same chapter, cited at ~9243). Zero boundary map splits off the top cohomology object.

Source reference. Original lemma 05QT: sources/stacks-derived/pinned-original-source.tex, source lines 917–917. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 08J5: sources/stacks-derived/pinned-original-source.tex, source lines 4293–4293. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let K be an object of $D^b(\mathcal{A})$ such that $\mathrm{Ext}^p_{\mathcal{A}}(H^i(K), H^j(K)) = 0$ for all $p \geq 2$ and $i > j$. Then K is isomorphic to the direct sum of its cohomologies: $K \cong \bigoplus H^i(K)[-i]$.*

Proof. Choose a, b such that $H^i(K) = 0$ for $i \notin [a, b]$. We will prove the lemma by induction on $b - a$. If $b - a \leq 0$, then the result is clear. If $b - a > 0$, then we look at the distinguished triangle of truncations

$$\tau_{\leq b-1}K \to K \to H^b(K)[-b] \to (\tau_{\leq b-1}K)[1]$$

see Remark 08J5. By Lemma 05QT if the last arrow is zero, then $K \cong \tau_{\leq b-1}K \oplus H^b(K)[-b]$ and we win by induction. Again using induction we see that

$$\mathrm{Hom}_{D(\mathcal{A})}(H^b(K)[-b], (\tau_{\leq b-1}K)[1]) = \bigoplus\nolimits_{i<b} \mathrm{Ext}^{b-i+1}_{\mathcal{A}}(H^b(K), H^i(K))$$

By assumption the direct sum is zero and the proof is complete. □

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